Ingest pit_stops.json file

```
dbutils.widgets.text("p_data_source", "")
v_data_source = dbutils.widgets.get("p_data_source")
```

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

Step 1 - Read the JSON file using the spark dataframe reader API

```
from pyspark.sql.types import StructType, StructField, IntegerType, StringType
```

```
lap_times_schema = StructType(fields=[StructField("raceId", IntegerType(), False),
                                      StructField("driverId", IntegerType(), True),
                                      StructField("lap", IntegerType(), True),
                                      StructField("position", IntegerType(), True),
                                      StructField("time", StringType(), True),
                                      StructField("milliseconds", IntegerType(), True)
                                     ])
```

```
lap_times_df = spark.read \
.schema(lap_times_schema) \
.csv(f"{raw_folder_path}/{v_file_date}/lap_times")
```

```
```

```
display(lap_times_df)
```

Table

	raceId	driverId	lap	position	time	milliseconds
112	1053	1	49	5	1:19.154	79154
113	1053	1	50	4	1:17.858	77858
114	1053	1	51	4	1:18.512	78512
115	1053	1	52	4	1:19.001	79001
116	1053	1	53	4	1:18.805	78805
117	1053	1	54	4	1:19.206	79206
118	1053	1	55	3	1:17.950	77950
119	1053	1	56	3	1:18.505	78505
120	1053	1	57	3	1:18.622	78622
121	1053	1	58	3	1:18.957	78957
122	1053	1	59	3	1:18.534	78534
123	1053	1	60	2	1:16.702	76702
124	1053	1	61	2	1:18.824	78824
125	1053	1	62	2	1:18.787	78787
126	1053	1	63	2	1:18.038	78038
127	1053	844	1	3	1:47.198	107198

1,124 rows

Step 2 - Rename columns and add new columns

- Rename driverId and raceId
- Add ingestion_date with current timestamp

```
lap_times_with_ingestion_date_df = add_ingestion_date(lap_times_df)
```

```
from pyspark.sql.functions import current_timestamp, lit
```

```
final_df = lap_times_df.withColumnRenamed("driverId", "driver_id") \
.withColumnRenamed("raceId", "race_id") \
.withColumn("ingestion_date", current_timestamp()) \
.withColumn("data_source", lit(v_data_source)) \
.withColumn("file_date", lit(v_file_date))
```

```
display(final_df)
```

Table

	race_id	driver_id	lap	position	time	milliseconds	ingestion_date
1	1053	830	1	1	1:38.603	98603	2024-07-24T13:51:50.109+00:...
2	1053	830	2	1	2:29.163	149163	2024-07-24T13:51:50.109+00:...
3	1053	830	3	1	2:23.247	143247	2024-07-24T13:51:50.109+00:...
4	1053	830	4	1	2:20.332	140332	2024-07-24T13:51:50.109+00:...
5	1053	830	5	1	2:25.691	145691	2024-07-24T13:51:50.109+00:...
6	1053	830	6	1	2:20.804	140804	2024-07-24T13:51:50.109+00:...
7	1053	830	7	1	1:36.303	96303	2024-07-24T13:51:50.109+00:...
8	1053	830	8	1	1:32.925	92925	2024-07-24T13:51:50.109+00:...
9	1053	830	9	1	1:30.953	90953	2024-07-24T13:51:50.109+00:...
10	1053	830	10	1	1:30.130	90130	2024-07-24T13:51:50.109+00:...

12	1053	830	12	1	1:28.814	88814	2024-07-24T13:51:50.109+00:...
13	1053	830	13	1	1:28.119	88119	2024-07-24T13:51:50.109+00:...
14	1053	830	14	1	1:27.740	87740	2024-07-24T13:51:50.109+00:...

1,124 rows

Step 3 - Write to output to processed container in parquet format

```
# final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_processed.lap_times")
```

```
# overwrite_partition(final_df, 'f1_processed', 'lap_times', 'race_id')
```

```
merge_condition = "tgt.race_id = src.race_id AND tgt.driver_id = src.driver_id AND tgt.lap = src.lap AND tgt.race_id = src.race_id"
merge_delta_data(final_df, 'f1_processed', 'lap_times', processed_folder_path, merge_condition, 'race_id')
```

```
# display(spark.read.parquet(f"{processed_folder_path}/lap_times"))
```

```
%sql
select * from f1_processed.lap_times;
```

Table

	race_id	driver_id	lap	position	time	milliseconds	ingestion_date	
1	847	20	1	1	2:18.174	138174	2024-07-24T13:50:26.497+00:...	
2	847	20	2	1	2:06.919	126919	2024-07-24T13:50:26.497+00:...	
3	847	20	3	1	2:05.303	125303	2024-07-24T13:50:26.497+00:...	
4	847	20	4	1	2:05.715	125715	2024-07-24T13:50:26.497+00:...	
5	847	20	5	1	1:36.175	96175	2024-07-24T13:50:26.497+00:...	
6	847	20	6	1	1:34.827	94827	2024-07-24T13:50:26.497+00:...	
7	847	20	7	1	1:35.452	95452	2024-07-24T13:50:26.497+00:...	
8	847	20	8	1	1:41.724	101724	2024-07-24T13:50:26.497+00:...	
9	847	20	9	1	2:13.172	133172	2024-07-24T13:50:26.497+00:...	
10	847	20	10	1	2:08.183	128183	2024-07-24T13:50:26.497+00:...	
11	847	20	11	1	2:07.373	127373	2024-07-24T13:50:26.497+00:...	
12	847	20	12	1	2:01.938	121938	2024-07-24T13:50:26.497+00:...	
13	847	20	13	1	1:32.888	92888	2024-07-24T13:50:26.497+00:...	
14	847	20	14	1	1:33.332	93332	2024-07-24T13:50:26.497+00:...	

10,000+ rows | Truncated data due to row limit

```
Success
```